MIL-P-20664C
9 February 1977
SUPERSEDING
MIL-P-20664B
14 September 1967

MILITARY SPECIFICATION

PLANIMETER, POLAR, COMPENSATING, WITH CASE

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers compensating polar planimeters which are utilized for measuring areas of a plane figure by tracing or optically observing its boundary lines.

2. APPLICABLE DOCUMENTS

2.1 Issues of documents. The following documents of the issue in effect on date of invitation for bids or request for proposal form a part of this specification to the extent specified herein:

SPECIFICATIONS

FEDERAL

QQ-S-781	- Strapping, Steel, and Seals.
PPP-B-601	- Boxes, Wood, Cleated-Plywood.
PPP-B-636	- Boxes, Shipping, Fiberboard.

MILITARY

MIL-P-116	- Preservation-Packaging, Methods of.

FSC 6675

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: US Army Mobility Equipment Research and Development Command, ATTN: DRDME-DS Fort Belvoir, VA 22060 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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STANDARDS

MILITARY

MIL-STD-105	- Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129	- Marking for Shipment and Storage.
MIL-STD-130	- Identification Marking of US Military Property.

(Copies of specification and standards required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

NATIONAL MOTOR FREIGHT TRAFFIC ASSOCIATION, INC., AGENT

National Motor Freight Classification.

(Application for copies should be addressed to the American Trucking Associations, Inc., ATTN: Traffic Department, 1616 P Street, NW, Washington, DC 20036.)

UNIFORM CLASSIFICATION COMMITTEE, AGENT

Uniform Freight Classification.

(Application for copies should be addressed to the Uniform Classification Committee, ATTN: Tariff Publishing Officer, Room 1106, 222 South Riverside Plaza, Chicago, IL 60606.)

(Technical society and technical association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

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3. REQUIREMENTS

3.1 Description. The planimeter shall be of the compensating polar type. The planimeter shall consist of two principal parts: The tracer arm which carries the tracing point and recording carriage with measuring wheel; and, the polearm attached to the pole around which the instrument revolves. The polearm shall be joined to the recording carriage by means of a separable ball-type point or optical tracer so that diagrams can be measured with the polearm on the right and then again on the left of the tracer arm, and by taking the mean of the two readings all instrument errors are compensated. It shall provide readings in hundredths of units of area, in inches or centimeters, depending upon the adjustable position of the recording carriage on the tracer arm.

3.2 Material. Material shall be corrosion resistant as specified herein. Materials not specified shall be selected by the contractor and shall be subject to all provisions of this specification.

3.3 Tracer arm. The tracer arm shall be provided with a tracer needle or optical tracer, as specified (see 6.2). The arm shall be graduated for an effective length of not less than 6 inches to enable setting the recording carriage vernier to hundredths at any scale within the range of the instrument. The tracer needle shall have an adjustable point. The tracer needle or optical tracer shall be provided with a fingertip guider positioned at the outer end of the tracer arm.

3.4 Polearm. The polearm shall be adjustable from approximately 6 inches to 13 inches and shall have an engraved scale and vernier for settings so that when the instrument is used with the pole inside of a figure, the instrument constant can be adjusted to a round number (20,000) for any setting of the recording carriage. A screw clamp shall lock the polearm at any desired setting. The parts of the polearm shall fit snugly and slide freely to permit quick settings. The arm shall have an anchor weight at one end, fitted with a high-carbon tool steel needle on the underside to prevent slipping on the working surface. The weight shall be of corrosion-resistant material, and shall be attached to the polearm. The other end of the polearm shall be fitted with a corrosion-resistant ball-end pin for fitting into a socket in the recording carriage to form a ball-and-socket joint.

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3.5 Recording mechanism. The recording mechanism shall consist of a measuring wheel with vernier, a recording disk for counting the revolutions of the measuring wheel, a support wheel on the opposite side of the tracer arm with its axis at right angles thereto when these parts are in correct position on the assembled planimeter, and a supporting carriage for the mechanism. The graduated parts of the measuring wheel and vernier shall be white with black graduations and numbering. The wheel shall be divided into 100 parts with every tenth part numbered, and the vernier used in conjunction with the wheel shall permit one-thousandth of a revolution to be read. The measuring wheel, bearings, and shaft shall be of corrosion-resistant material. A vernier shall be attached to the upper part of the carriage and shall be designed and located to be used in conjunction with the graduations on the tracer arm to facilitate settings of the instrument to hundredths. A clamp and slow-motion screw shall be provided to facilitate accurate setting of the recording mechanism along the tracer arm. When the carriage is unclamped, it shall slide freely on the tracer arm, and all clamping and slow-motion screws shall work smoothly without binding. The measuring and support wheels shall turn freely with no tendency to bind.

3.6 Graduations. The graduations on the tracer arm and polearm shall be machine divided and shall be spaced no greater than 0.5 millimeter (mm) when the metric system is used, or 0.025 inch when the decimal inch system is used, shall be straight, uniform, parallel, permanently marked, and free from breaks or ragged edges, and shall be clear and distinct so as to be quickly and easily read. Graduation lines of equal lengths shall indicate aliquot parts of the scale. Lines indicating subdivisions of larger parts shall be shorter than the lines indicating major divisions so that the difference is distinguishable at a glance. Vernier scales shall read to one-tenth of corresponding least scale divisions. The error between any two graduation marks shall not exceed 0.002 inch.

3.7 Area coverage. The planimeter shall measure the area within a circle having a 16-inch radius when the pole is within the circle and shall measure the area of a circle 12 inches in diameter when the pole is outside of the circle.

3.8 Testing rule. A metal testing rule shall be furnished with each planimeter. The rule shall have an engraved indexing mark at one end and a 1/16-inch positioning needle projecting from the side opposite the indexing mark at the other end. Two holes shall be drilled to snugly fit the tracing point of the planimeter on a line connecting

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the index mark and the positioning needle. The holes on the testing rule shall be at positions which will give 10 square inches and 100 square centimeters of area. The areas of the circles enscribed by rotating the drilled hole about the needle point as a center shall be engraved on the rule beside their corresponding hole, and the numerical value shall be within plus or minus 1 percent of the area computed by using the spacing between the positioning needle and the center of the drilled hole as the radius of a circle.

3.9 Chart. A chart shall be prepared and furnished for use in the conduct of the tests specified in 4.4.2.3 and for inclusion as part of the maintenance and operation manual specified in 6.3. The chart shall contain two continuous line graphs relating the position of the recording carriage on the tracer arm to the value of the vernier unit on the measuring roller. One of the graphs shall give the relationship in terms of square inches. The other graph shall give the relationship in terms of square centimeters. The graphs shall cover minimum ranges for vernier unit values of 565 to 2032 for square inches and 875 to 3150 for square centimeters.

3.10 Accuracy. The instrument shall check the areas engraved on the testing rule and vernier units indicated on the graphs within 1.0 percent.

3.11 Case. The planimeter shall be furnished in a fitted case which shall be lined with a material which is fungus resistant or treated to be fungus resistant. The case shall have a hinged cover and a fastener. The material used to fabricate the case shall be fungus resistant or treated to be fungus resistant.

3.12 Identification marking. The planimeter and case shall be identified in accordance with MIL-STD-130.

3.13 Workmanship. All parts of the planimeter shall be clean and free from sand, dirt, fins, sprues, scale, flux, or other harmful extraneous material. All edges shall be rounded and chamfered.

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4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Material inspection. The contractor is responsible for insuring that materials used are manufactured, examined, and tested in accordance with referenced specifications and standards, as applicable.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

(a) Quality conformance inspection (see 4.3).
(b) Inspection of packaging (see 4.5).

4.3 Quality conformance inspection.

4.3.1 Sampling. Sampling for examination and tests shall be in accordance with MIL-STD-105.

4.3.2 Examination. Samples selected in accordance with 4.3.1 shall be examined as specified in 4.4.1. AQL shall be 2.5 percent defective.

4.3.3 Tests. Samples selected in accordance with 4.3.1 shall be tested as specified in 4.4.2.1 through 4.4.2.3. AQL shall be 2.5 percent defective.

4.4 Inspection procedure.

4.4.1 Examination. The planimeters shall be examined as specified herein for the following major defects:

101. Materials not as specified.
102. Recording mechanism, when unlocked, does not slide freely on the tracer arm.
103. Clamping and slow-motion screws bind or do not turn freely.

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104. Measuring and support wheels do not turn freely.
105. Graduations not as specified.
106. Polearm not adjustable within specified limits.
107. Identification marking not as specified.
108. Workmanship not as specified.

4.4.2 Tests.

4.4.2.1 Area coverage. Set up the planimeter at maximum scale-setting range and move the tracing point so as to describe the maximum circle. Inability of the tracing point to describe a circle having a minimum radius of 16 inches shall constitute failure of this test.

4.4.2.2 Testing-rule accuracy. Check the area value engraved on the rule of the planimeter by measuring the spacing between the needle point and the center of the drilled hole to within 0.01 inch for the inch area and 0.05 centimeter for the metric area and compute the area of the circle which would result from using this spacing as a radius. Nonconformance to 3.8 shall constitute failure of this test.

4.4.2.3 Area-measuring accuracy. Check the planimeter for area-measuring accuracy as follows:

(a) The areas engraved on the testing rule:
  (1) The 10-square-inch value. By means of the square inches graph (see 3.9), locate the 1000 vernier unit value and its corresponding tracer arm setting for positioning the recording carriage. Set the recording carriage at the tracer arm setting, insert the needle point of the test rule into the drafting surface, and place the tracing arm point of the planimeter into the hole in the testing rule having 10 square inches engraved beside it. Mark on the drafting surface the beginning as shown by the indexing mark on the rule and then rotate the rule about the impressed point as a center; stop at the beginning mark after completing 360 degrees. Record the number of vernier units required for one revolution of the test rule. Retaining the same setting of the recording carriage on the tracer arm, rotate the rule two more times around the impressed point, each time recording the number of vernier units required for one revolution of the test rule.

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Compute the average of the three revolutions. Multiply the average value by the necessary constant to convert the reading to square inches. An average value not within plus or minus 1.0 percent of 10 square inches shall constitute failure of this test.

(2) The 100-square-centimeter value. By means of the square centimeters graph (see 3.9), locate the 1000 vernier unit value and its corresponding tracer arm setting for positioning the recording carriage. Repeat the foregoing procedure (a) (1), but use the hole on the testing rule for centimeters and convert the average value to square centimeters. An average value not within 1.0 percent of 100 square centimeters shall constitute failure of this test.

(b) Vernier units on the graphs:

(1) Use the 10-square-inch hole on the test rule. By means of the square inches graph (see 3.9), select two widely spaced vernier units (one on each side of the 1000) and their corresponding tracer arm setting for positioning the recording carriage. Using either one of the selected vernier units, set the recording carriage at the corresponding tracer arm setting, insert the needle point of the test rule into the drafting surface, and place the tracing arm point of the planimeter into the hole in the testing rule having 10 square inches engraved beside it. Mark on the drafting surface the beginning as shown by the indexing mark on the rule and then rotate the rule about the impressed point as a center; stop at the beginning mark after completing 360 degrees. Record the number of vernier units required for one revolution of the test rule. Retaining the same setting of the recording carriage on the tracer arm, rotate the rule two more times around the impressed point, each time recording the number of vernier units required for one revolution of the test rule. Compute the average of the three revolutions. An average value not within 1.0 percent of the vernier unit value being used shall constitute failure of this test. Repeat the entire foregoing procedure for the second selected vernier unit. This average value shall check the second vernier unit value being used within plus or minus 1.0 percent. An average value not within 1.0 percent of the vernier unit value being used shall constitute failure of this test.

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(2) Use the 100-square-centimeter hole on the test rule. By means of the square centimeters graph (see 3.9), select two widely spaced vernier units (one on each side of the 1000) and their corresponding tracer arm setting for positioning the recording carriage. Using either one of the selected vernier units, set the recording carriage at the corresponding tracer arm setting, insert the needle point of the test rule into the drafting surface, and place the tracing arm point of the planimeter into the hole in the testing rule having 100 square centimeters engraved beside it. Mark on the drafting surface the beginning as shown by the indexing mark on the rule and then rotate the rule about the impressed point as a center; stop at the beginning mark after completing 360 degrees. Record the number of vernier units required for one revolution of the test rule. Retaining the same setting of the recording carriage on the tracer arm, rotate the rule two more times around the impressed point, each time recording the number of vernier units required for one revolution of the test rule. Compute the average of the three revolutions. An average value not within 1.0 percent of the vernier unit value being used shall constitute failure of this test. Repeat the entire foregoing procedure for the second selected vernier unit. This average value shall check the second vernier unit value being used within plus or minus 1.0 percent. An average value not within 1.0 percent of the vernier unit value being used shall constitute failure of this test.

4.5 Inspection of packaging.

4.5.1 Quality conformance inspection of pack.

4.5.1.1 Unit of product. For the purpose of inspection, a completed pack prepared for shipment shall be considered a unit of product.

4.5.1.2 Sampling. Sampling for examination shall be in accordance with MIL-STD-105.

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4.5.1.3 Examination. Samples selected in accordance with 4.5.1.2 shall be examined for the following major defects. AQL shall be 2.5 percent defective.

109.	Materials, methods, or containers not as specified.	Each incorrect material, method, or container shall be considered one defect.
110.	Strapping not as specified.	
111.	Marking illegible, incomplete, incorrect, or missing.	

5. PACKAGING

5.1 Preservation. Preservation shall be Level A or Commercial, as specified (see 6.2).

5.1.1 Level A. Each complete planimeter with testing rule and data publications shall be preserved in accordance with MIL-P-116, Method IA-8.

5.1.1.1 Consolidation. Planimeters preserved in accordance with 5.1.1 shall be consolidated in close-fitting boxes conforming to PPP-B-636, W6c, style optional. Strapping of boxes shall not be required.

5.1.1.2 Repair parts and tools. Repair parts and tools shall be preserved in accordance with the criteria and the applicable method of preservation of MIL-P-116.

5.1.2 Commercial. Planimeters with technical publications, repair parts, and tools shall be preserved to afford protection against deterioration and damage during shipment and handling from the contractor to the initial destination.

5.2 Packing. Packing shall be Level A, Level B, or Commercial, as specified (see 6.2).

5.2.1 Level A. Planimeters preserved as specified in 5.1 shall be packed in close-fitting boxes conforming to PPP-B-601, Overseas Type, style optional. Strapping shall conform to QQ-S-781, Class 1, Type I or Type IV, size as applicable. Unless otherwise specified (see 6.2), strapping shall be Finish B. When specified (see 6.2), strapping shall be Finish A.

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5.2.2 Level B. Packing shall be as specified for Level A, except that boxes shall be Domestic Type and strapping shall be Finish A.

5.2.3 Commercial. Planimeters, with technical publications, repair parts, and tools shall be packed to assure carrier acceptance and safe delivery to destination at the lowest ratings of the carrier in compliance with Uniform Freight Classification rules or National Motor Freight Classification rules.

5.3 Marking.

5.3.1 Military packaging. Marking shall be in accordance with MIL-STD-129.

5.3.2 Commercial packaging. Commercial packaging marking shall reflect as a minimum, the item name, national stock number, quantity, contract or purchase order number, and appropriate address markings. The marking shall be applied with a waterproof material, shall be legible, and shall provide a definite contrast.

6. NOTES

6.1 Intended use. The planimeter is intended to furnish the most efficient means of determining the area of figures with irregular or curved boundaries.

6.2 Ordering data. Procurement documents should specify the following:

(a) Title, number, and date of this specification.
(b) Type tracer required (see 3.3).
(c) Degree of preservation and degree of packing required (see 5.1 and 5.2).
(d) When other than Finish B strapping is required (see 5.2.1).

6.3 Data requirements. The contracting officer should include requirements for such data as technical publications, instructional materials, illustrated parts lists, and contractor's maintenance and operation manual to be furnished with each planimeter. The manual shall include the chart specified in 3.9.

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6.4 Provisioning. The contracting officer should include provisioning requirements for repair parts and maintenance tools as necessary (including any special tools), and instructions regarding shipment of planimeters.

Custodians:
Army - ME
Air Force - 99

Review activity:
DSA - GS

User activities:
Army - ~~EL~~ ER
Navy - YD

Preparing activity:
Army - ME

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DOCUMENT IDENTIFIER *(Number)* AND TITLE
MIL-P-20664C Planimeter, Polar, Compensating, with Case

NAME OF ORGANIZATION AND ADDRESS OF SUBMITTER

☐ VENDOR ☐ USER ☐ MANUFACTURER

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